



Exploratory and Explanatory Analysis Transcript

Hello everyone, welcome to BITS PLANET DIGITAL, today we will be continuing our learning journey as part of our data visualisation and storytelling course. We will take you through two important concepts today called exploratory analysis and explanatory analysis and also context of visualising data. These two concepts I would say are philosophical approaches in understanding and building visualisations.

This would better help you to build visualisations that can effectively communicate the content across a wide variety of audience to the particular purposes that you are intending to transfer the knowledge or information and it will help them better understand the content that you are trying to present. I would say this is more of a philosophical or a theoretical approach towards designing visuals. This is not ground hard fast rules that you are learning in this particular content, but it is more of an approach that is designed for delivering efficiency.

We will now see the difference between explanatory analysis and exploratory analysis. Whenever we begin an analysis especially for visualising data, we have to understand the composition of the data, what is its structure, where is it coming from, what are the different types of variables in it. All this would form under exploratory analysis which means you are exploring the data, data structure, components, variables, their nature, how are they distributed, what is the nature of those variables, whether there are numerical variables or their textual characters, whether the data is complete, all these features.

This would give you a very good command on the data structure, underlying data structure that you can use for your analysis and also it also in some sense sets the boundaries with which you can carry out your analysis. So, exploratory analysis is first step for any data visualisation process. In the exploratory analysis, your main aim is to filter, mine and extract useful insights that you can later build into the visualisation and use them for explaining the phenomenon to your audience.

Now comes the explanatory part. Explanatory analysis is built on exploratory analysis where you try to explain the insights that you found during the exploratory analysis to your audience driving home a particular point. Let us go back to the example that we have taken from the last class, the electoral voter participation in the 2024 Lok Sabha elections.

So, once we started to collect the data, I would like you to refer to the 7 stages of data visualisation which is acquire, parse, filter, mine, represent, refine and interact. These are the 7 steps of data visualisation. We start with acquiring the data and we have seen we have got the data in the tabular format that is on the left side.

So, you are doing the exploratory analysis at this stage where you are trying to understand the data structure, you know the data is organised as per the states and the key variable that we are looking here is at voter turnout ratio which is already calculated in the excel sheet.



Sometimes you might not have this variable readily available, you might have to construct this variables. So, exploratory analysis is the step where you try to understand what is your main variable of interest, whether it is readily available and do some calculations and also you look for the data integrity in this process, whether there are any missing values, number of outliers, the distribution of different variables.

While we go through the course, we will teach you the techniques and tools to do the exploratory data analysis which is also called as EDA and there are python suits softwares that are available that readily do this EDA and help you in this process and we will be taking them in the experiential learning part. However, to just to get the concepts across in the exploratory analysis part, you are trying to understand the data and see what it is speaking to you and all the steps that you try to do to get insights would fall under exploratory analysis. However, on the right hand side, you have the final visualisation that we showcased in the last class showing the voter turnout ratio across states with the average voter participation highest state and the lowest state with respect to voter turnout participation.

This is the visual that you would use to explain to your audience the nature of voter turnout participation across Indian states. This is where the exploratory feature is coming up using the same visual, you are trying to talk to the audience conveying a particular point. Now, coming to exploratory analysis, the main feature of exploratory analysis is asking questions.

What is the data trying to tell me? What is the relationship between different variables that are there in the data? And most importantly, do I have sufficient diversity in my data to test various relationship among the variables to build a story or to get a trend that is there to showcase that this is what is happening between two variables or a variable x is causing variable y after controlling for certain features and these are the additional features that I have to control if I have to get the pure effect of x on y . All these questions can be answered only if you have sufficient variety in the data. If you have very simple data, naturally the analysis that you can do with them is also limited. So, it is advised to have a diversified data structure or data to start with to build visuals and you can refine them as you go to tailor the particular visual to give out the output that you intend to convey.

However, if you start with very sparse data, the depth of your analysis is going to be limited. So, this is something as data analysts or business analysts, we should try to collect the widest possible data and refine it as we go. The final conclusion from this slide is looking for insights.

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